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Detection of alcohol-involved driving via machine learning using eye-tracking and vehicle data

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ABSTRACT

Introduction: Alcohol impaired driving remains a significant public safety problem, prompting the United States and the EU to move towards the inclusion of impaired driving prevention technology in future vehicle production. A large body of literature suggests acute alcohol use produces changes in ocular behavior and vehicle control, suggesting driver monitoring system (DMS) data can be used to detect alcohol-impaired driving. *Method:* We use DMS data from a simulated driving study to assess the role of different monitoring lengths, data inputs, and machine learning methods on the accurate identification of alcohol presence using a baseline-controlled study with no alcohol at baseline, followed by four drives after alcohol dosing with BrACs of 0.10, 0.085, 0.07, and 0.055 g/210 L. *Results:* Using data from an 8-minute segment of rural driving, we find convolution-based feature extraction methods using vehicle and ocular inputs can differentiate between samples of sober and alcohol-dosed driving with a high degree of accuracy (AUC = 0.955), outperforming models using only ocular inputs (AUC = 0.916) and only vehicle inputs (AUC = 0.880). When applied to new driving environments, models involving ocular features continue to perform well (AUC > 0.95), while models using only vehicle-based features are less predictive. We find pupil and blink behavior are the most influential predictors of alcohol presence. Future research should consider a broader range of BrACs and fatigue combinations, particularly if a DMS aims to differentiate between alcohol-impaired and fatigued states. *Practical Application:* As policymakers explore the feasibility of DMS to prevent alcohol-involved accidents, we find these systems should focus on longer measurement durations and utilize both vehicle and eye data (especially blink and pupil behavior) to achieve optimal performance. Furthermore, we find that machine learning models trained to detect patterns in eye data can be successfully applied in new driving environments without retraining.

1. Introduction

In the United States, nearly 12,000 people were killed in vehicular crashes where at least one driver was alcohol-impaired in 2024. Despite long-term decreases in the total number of annual crash fatalities, the percentage of fatalities involving alcohol has remained relatively constant at approximately 30% since the 1990s (National Highway Traffic Safety Administration [NHTSA], 2026). In 2023, the NHTSA published an *Advanced Notice of Proposed Rule Making* aiming to set standards for new technologies used to prevent impaired driving (National Highway Traffic Safety Administration, 2023), sparking increased research attention in the passive detection of alcohol impaired driving using driver monitoring systems (DMS).

Underpinning the development of passive alcohol detection systems is a large body of research on the changes alcohol and other drugs induce in psychomotor control and ocular behavior. Numerous studies have found acute alcohol use degrades driving performance, observing decreased lateral and longitudinal control, slower reaction times, and riskier decision making (Simmons et al., 2022). Acute alcohol use is also associated with irregular pupil shape due to constriction and dilation, delayed pupillary light reflex, reduced ability to track moving objects, and other changes in eye movement patterns (Silva et al., 2017; Tivesten

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Table 1
Categories and descriptions of all engineered features used by random forest and XGBoost machine learning models. Full summary statistics include mean, standard deviation, 5th and 95th percentiles, skewness, kurtosis, and middle 90th percentile range.

Feature group	Feature subgroup	Measures	Variable Name	Description
Eye	Eyelid behavior	Proportions and Counts	Left PERCLOS	Proportion of time left eyelid closed > 80%
			Right PERCLOS	Proportion of time right eyelid closed > 80%
			Blinks	Number of blinks detected
		Full summary statistics	Blink Duration	Duration of blinks
	Pupil characteristics	Full summary statistics	Left Pupil Diameter	Left pupil diameter (mm)
		Full summary statistics	Right Pupil Diameter	Right pupil diameter (mm)
	Head position	Full summary statistics	Head Position X	Head position side-to-side
			Head Position Y	Head position up-down
			Head Position Z	Head position front-back
	Gaze direction	Proportions	PERLOOK	Proportion of time eyes off road
		Full summary statistics	Left Gaze X	X-coordinate left eye gaze vector
			Left Gaze Y	Y-coordinate left eye gaze vector
			Left Gaze Z	Z-coordinate left eye gaze vector
			Right Gaze X	X-coordinate right eye gaze vector
			Right Gaze Y	Y-coordinate right eye gaze vector
			Right Gaze Z	Z-coordinate right eye gaze vector
			Unified Gaze Pitch	Both eyes gaze pitch angle (deg)
			Unified Gaze Yaw	Both eyes yaw angle (deg)
	Saccade & fixation	Mean	Fixations	Number of fixations
			Fixation Duration	Duration of fixations
		Mean & Std dev	Horizontal Dispersion	Horizontal spread of gaze
			Vertical Dispersion	Vertical spread of gaze
			Peak Velocity Horizontal	Max horizontal saccade velocity
			Peak Velocity Vertical	Max vertical saccade velocity
Vehicle	Longitudinal control	Full summary statistics	Vehicle Speed	Speed of vehicle
			Brake Force	Force on brake pedal
			Acceleration Pedal	Accelerator pedal position
	Lateral control	Full summary statistics	Steering Angle Rate	Rate of steering wheel angle change
			Lane Deviation	Deviation from lane center
			Steering Angle	Steering wheel angle

et al., 2023; Maurage et al., 2020). Taken together, these findings suggest that DMS using a combination of ocular and vehicular data may be able to accurately identify instances of alcohol-impaired driving. However, statistical models typically enforce strict functional relationships and may exhibit underwhelming predictive performance, thus prompting the use of complex machine learning methods to assess the predictive value of these associations.

Several previous studies have explored the use of machine learning methods to detect alcohol-impaired driving using ocular and vehicle data, most of which using driving simulators (Koch et al., 2023), though some used real vehicles on test tracks (Deuber et al., 2025). Additional examples include Çetinkaya and Acarman (2023), who used tree-based feature selection and gradient boosting to classify drivers as impaired using eye glance behaviors and data from vehicle sensors. Related work, focusing on identifying driver fatigue, includes McDonald et al. (2014) who used the random forest algorithm and steering wheel angle data to detect instances of drowsy driving, and Dong et al. (2022) who used non-intrusive image sequences of a driver's face to detect fatigued and distracted driving via random forest and neural network models.

The scope of this existing work lags the rapid development of camera-based DMS technologies. As of 2024, NHTSA reported more than 100 camera-based systems currently under development. However, despite significant governmental and industry interest, relatively little data has been published on the performance of camera-based alcohol-detection systems (Prendez, 2024). Our work builds upon existing research into the passive detection of alcohol presence using DMS by evaluating the detection performance of several modern machine learning approaches developed over varying time intervals, ranging from 15 to 90 s, using data from a high-fidelity driving simulator equipped with an automotive-grade DMS that recorded eye and facial features including head position, gaze direction, eye closure, and pupil measurements. The driving simulator platform recorded driver inputs and vehicle states, thereby allowing us to assess the relative contributions of ocular and vehicular information in the accurate detection of alcohol presence. We also examine the effectiveness of machine learning models that were trained to detect alcohol presence in a specific driving environment when applied to data from new environments, and we explore strategies for maintaining high detection rates while limiting the number of false positives.

2. Methods

2.1. Data collection

Subjects recruited using a large email registry and word of mouth completed a single visit driving simulator study consisting of a baseline (sober) drive, alcohol dosing, and four post-dose follow-up drives. Exclusion criteria included pregnancy, recent drug use, physical or psychiatric conditions, and a history of motion sickness. Simulated drives were performed using a Quarter Cab miniSim, PC-based driving simulator featuring three 48-inch, high-definition (1080p) LCD displays capable of providing a comprehensive forward field of vision (National Advanced Driving Simulator, 2026). Previous research has validated miniSim's sensitivity to alcohol impairment and correspondence with driving performance on real roadways (Kay et al., 2013).

An automotive-grade DMS produced by Seeing Machines Ltd (Seeing Machines Ltd, 2026) was integrated into the simulator, allowing for the simultaneous collection of ocular and vehicular data, which were processed and aligned to produce multivariable 60 Hz time-series data spanning the duration of each drive. The DMS uses near-infrared lighting and a camera focused on the driver's face to capture ocular parameters, which are processed to determine head position, gaze direction, and eye closure. Each participant's facial shape was matched with a preset shape to further facilitate accurate head and eye-tracking. Each simulated drive had an approximate duration of 25-minutes and featured a mixture of driving environments, including straight and

curved rural roadways with a 55 mile per hour speed limit in a nighttime environment with varying amounts of traffic, as well as a stretch of gravel roadway with gentle curves, no lane marking, and no posted speed limit. Each drive used a randomly assigned scenario order and route variation to minimize learning effects.

Alcohol doses were calculated using the Sahlgrenska formula with a peak breath alcohol concentration (BrAC) of 0.115 g/210 L. The calculated dose was mixed with orange juice and divided evenly across three drinks, which were consumed over consecutive 10-minute periods such that the entire dose was consumed within 30-minutes. BrAC was measured at 5-minute intervals during the dosing period using a calibrated Alco-Sensor FST testing instrument to ensure the target BrAC was reached for all participants. The four post-dose drives took place at BrAC concentrations of 0.10, 0.085, 0.07, and 0.055 g/210 L with an allowed margin of error of 0.01. BrAC was continually monitored, and subjects outside of the allowed margin or with levels declining faster than expected were given small, secondary doses of alcohol to bring them back to the target BrAC by the start of the next drive. Subjects also completed a brief questionnaire before and after each drive which included the Karolinska Sleepiness Scale (KSS) (Shahid et al., 2012), a validated measure of fatigue ranging from 0 to 9, as well as their self-reported readiness to drive (yes/no).

2.2. Data preparation

In total, 36 subjects completed the study, yielding 180 drives. Due to missing or corrupted eye or vehicle data, 27 drives were excluded, leaving 153 available for analysis. All models were trained using data recorded during a straight segment of a two-lane rural road encountered near the end of each simulated drive. This segment was approximately 8-minutes in duration, and we divided it into intervals using a sliding window approach that allowed 50% overlap between consecutive intervals. We repeated this procedure using interval lengths of 15, 30, 45, 60, 75, and 90 s to evaluate the influence of measurement duration on the accuracy of our models. To prevent data leakage arising from the overlapping nature of these intervals, all intervals from the same drive were randomly assigned to either a training or validation set, and the training-validation process was repeated 10 times using different random assignments to ensure our reported performance metrics are robust to chance variation introduced by the data splitting procedure.

2.3. Machine learning models

Classification models were trained to predict whether an interval was from the sober, baseline drive, or from an alcohol-dosed drive. We adopt two general approaches when developing models: *feature engineering* and *feature learning*. In the former approach, the multivariate time-series data recorded by the DMS within interval is reduced to a set of derived features informed by relationships described in the literature on alcohol impairment. Machine learning models are then trained on these derived features to learn patterns that are associated with alcohol-presence. A similar approach was taken by Tivesten et al. (2023), who also developed machine learning methods for the detection of alcohol-impaired driving. Within each interval, we calculate the mean, standard deviation, 5th and 95th percentiles, skewness, and kurtosis of each of our selected inputs. We also calculate several additional predictive features, including the number of blinks, saccades, and fixations, using the 'saccades' library in R (von der Malsburg, 2019). The library uses a velocity-based detection algorithm which has been demonstrated to be reliable on eye-tracking data with frequencies below 100 Hz (Engbert and Kliegl, 2003). In total, 116 eye-related features and 40 vehicle-related features were engineered from the time-series data provided by the DMS. A complete list of all engineered features can be found in Table 1.

Feature engineering reduces each interval to a single data-point without a time dimension such that it can be used as an input into a conventional machine learning model. We consider random forest (Breiman, 2001) and gradient boosted trees, as implemented in the 'XGBoost' library (Chen & Guestrin, 2016), when classifying intervals using engineered features. Both methods are ensembles of decision trees, which allows them to learn complex, non-linear relationships and interactions from their inputs. Furthermore, these methods allow for the calculation of gain-based feature importance metrics, which can be used to quantify the influence of specific input features on predictive performance, thereby allowing us to assess the relative contributions of various types of input data on each model's performance. Shapely additive explanations (SHAP) (Lundberg & Lee, 2017) were also used to further evaluate the contributions of specific engineered features on individual model predictions.

We use cross-validated grid searches to ensure a sensible set of hyperparameters were used in each application of the XGBoost and Random Forest models. For random forest models, this search optimized the tree depth, number of features selected at each decision split, and total number of trees. For XGBoost models, we optimized the learning rate, number of trees, tree depth, number of features considered at each split, and the amount of L1 regularization. The XGBoost and random forest algorithms do perform feature selection, so all 146 features described in Table 1 were used by these models.

For our second approach, feature learning, we apply the Random Convolutional Kernel Transform (ROCKET) first proposed by Dempster et al. (2020) using regularized logistic regression as the final classifier. ROCKET uses a large number, 10,000 in our application, of convolutional kernels with randomly generated weights to produce a collection of feature maps that capture temporal patterns in the input time-series. Each of these feature maps is summarized by its percentage of positive values (PPV) and absolute maximum value (AM), then the resulting set of PPVs and AMs are used as inputs into the final classifier. In contrast with our feature engineering approach, which relies upon handcrafted features we have control over, ROCKET learns predictive patterns on its own from the multivariate time-series inputs provided by the DMS. The data streams used in the multivariate input were brake force, acceleration, lateral position, speed, steering wheel angle rate of change, the x, y, and z coordinates of gaze direction and head position, left and right pupil diameter and eye closure, and DMS estimated confidence scores for gaze and eye values.

2.4. Model evaluation

Each of our trained models was used to calculate prediction scores on withheld validation samples. In practice, prediction scores must be mapped to a predicted alcohol-presence status using a fixed threshold value. Thus, we report two different performance metrics for our models: the area under the receiver operating characteristic curve (AUC), which provides a comprehensive summary of classification performance across a broad range of prediction score thresholds, and the true positive rate (TPR) achieved at the lowest prediction score threshold where exactly zero false positives were observed in validation samples under consideration. Unless otherwise noted, the AUC and TPR values we report are averaged across the 10 randomly chosen training-validation splits.

In addition to evaluating models trained using all available ocular and vehicle features present under a given scheme (feature engineering and feature learning), we also train and evaluate separate sets of models that were constrained to use only eye-inputs and only vehicle-inputs. This provides performance measures for each combination of modeling approach (random forest, XGBoost, and ROCKET), interval duration (15, 30, 45, 60, 75, and 90 s), and input data (combined eye and vehicle, eye-only, vehicle-only). We report gain-based feature importance measures for the top-performing feature engineering model in each configuration, including the individual features with the strongest contributions, as well as the aggregate contributions of various nested categories of features described in Table 1. Aggregate

Table 2
Demographic characteristics of the n = 36 subjects who completed the study.

Age	21–65 years (mean = 30.9, SD = 10.8)
Sex	Male = 50%, Female = 50%
Hispanic	Yes = 4 (11.1%)
	No = 32 (88.9%)
Race	Asian/Asian American = 6 (16.7%)
	Black/African American = 3 (8.3%)
	White = 23 (63.9%)
	More than one race = 2 (5.6%)
	Prefer not to answer = 2 (5.6%)
Skin tone—Fitzpatrick scale	Type 1 = 0 (0%)
	Type 2 = 25 (69.4%)
	Type 3 = 8 (22.2%)
	Type 4 = 4 (11.1%)
	Type 5 = 3 (8.3%)
	Type 6 = 1 (2.8%)

contributions are reported on a raw (absolute) basis as well as on standardized (per feature) basis to account for differing numbers of features within each category and subcategory.

While our primary evaluations use validation data from the rural straight driving scenario on which our models were trained, we also evaluate each model on unseen data from two different driving environments from other portions of each simulated drive. The first scenario, *dark*, is an unlit segment of two-lane rural road with the same road surface and speed limit (55 mph) as the rural straight scenario used in model training, but it differs through its inclusion of a significant curve. The second scenario, *gravel*, features a road surface with a lower coefficient of friction, gentle curves, and no posted speed limit.

Finally, since our primary analyses make classifications at the interval level, we explore methods for allowing adjacent intervals within the same drive to "vote" on the alcohol status of the driver, allowing for a single interval's prediction to be overwritten if enough adjacent intervals suggest a particular status. More specifically, we consider rolling groupings of three adjacent intervals and allow the block to vote using simple majority and unanimous schemes for varying prediction score thresholds, tracking the true positive and false positive rate for each scheme at each threshold.

3. Results

Demographic characteristics of the n = 36 individuals who completed the study are provided in Table 2. Overall, the sample had an average age of 30.9 years (range 21–65), contained equal numbers of males and females, and included individuals from a variety of racial and ethnic backgrounds with differing skin tones, a relevant consideration for eye-tracking technology.

3.1. Model comparisons

Fig. 1 displays the mean AUC values across validation sets by interval length and modeling approach. Performance tended to be best when models were trained on longer duration intervals, with most approaches exhibiting an upward trend as interval length increases, apart from ROCKET models trained using eye inputs, whose performance reaches a plateau at an interval duration of 60-seconds. For the feature engineering approaches (random forest and XGBoost), models trained using only vehicle inputs tend to outperform models trained using only ocular inputs for every interval duration; however, the opposite is true for ROCKET models, where models trained using eye-only features results in better performance than models trained with vehicle-only features. This suggests there are predictive patterns in the eye-data time-series that are not being effectively captured by the engineered features used by the random forest and XGBoost models.

Overall, the best classification performance is achieved by ROCKET models using both ocular and vehicle features measured on 90-second

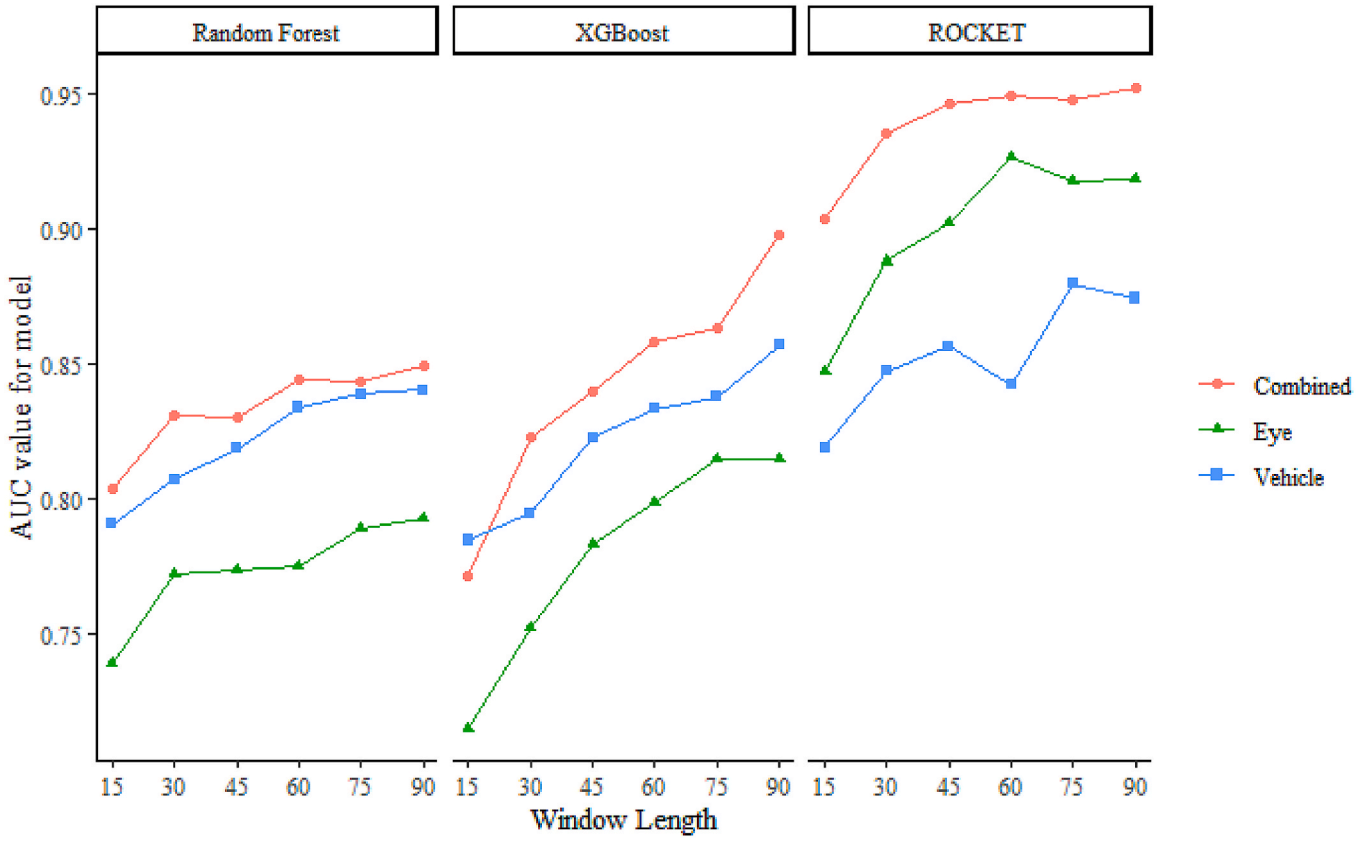


Fig. 1. Mean AUC across 10 randomly assigned validation sets for each combination of modeling approach, data type, and interval duration.

Table 3
Summary of AUC values across 10 randomly assigned validation sets for each modeling approach and data type trained using 90 s intervals.

Predictors	Model	Mean	Std Dev.	Min	Max
Eye	Random Forest	0.802	0.061	0.679	0.882
	XGBoost	0.833	0.051	0.741	0.898
	ROCKET	0.916	0.039	0.842	0.977
Vehicle	Random Forest	0.848	0.032	0.798	0.883
	XGBoost	0.865	0.039	0.788	0.902
	ROCKET	0.88	0.034	0.818	0.925
Combined	Random Forest	0.852	0.062	0.713	0.922
	XGBoost	0.902	0.038	0.83	0.952
	ROCKET	0.955	0.027	0.906	0.986

time intervals, which showed a mean AUC of 0.955 (sd = 0.039, range = 0.842–0.977). ROCKET was also the best performing method for the eye-only and vehicle-only sets of input features, achieving a mean AUC of 0.916 (sd = 0.039, range = 0.842–0.977) with the former and 0.880 (sd = 0.034, range = 0.818–0.925) with the latter. Table 3 provides summary results for all three modeling approaches using 90-second intervals.

Fig. 2 displays receiver operating characteristic (ROC) curves for each modeling approach using 30, 60, and 90 s intervals. While these curves support higher AUC values achieved by ROCKET, they show similar true positive rates at the portions of the curve where the false positive rate is near zero, with the higher AUC being largely due to better true positive rates achieved by ROCKET at intermediate thresholds where some false positives are tolerated. Fig. 3 further investigates this result, displaying the maximum TPR achieved by each modeling method using combined vehicle and eye data across each of the 10 randomly allocated validation splits at the prediction score threshold that produced exactly zero false positive classifications.

Despite the superior AUC scores achieved by ROCKET models, XGBoost models tend to achieve higher TPRs when compared to their ROCKET counterparts with the same interval duration. Altogether, the best performing XGBoost models, which used 90-second intervals, suggest TPRs near 60% can be achieved without making many false positive classifications using handcrafted eye and vehicle features.

3.2. Voting schemes

Fig. 4 displays the TPR achieved at varying classification thresholds using majority and unanimous voting schemes in comparison to regular classification where each interval only influences its own predicted class. The dashed lines annotate the lowest threshold at which exactly zero false positives are observed. For XGBoost models (left panels), we see that both majority and unanimous voting allow for a higher TPR to be achieved without making any false positive classifications. For ROCKET models (right panels), there is no meaningful improvement in TPR when implementing voting and targeting exactly zero false positives.

3.3. Feature importance

In contrast with ROCKET models, it is straightforward to assess the relative contributions of input features in XGBoost models using gain-based importance. Fig. 5 displays the total share of gain-based importance attributed to features derived from ocular and vehicle inputs. Because we used nearly three times as many ocular inputs (116 versus 40), the left panel displays raw, unadjusted feature importance shares, while the right panel standardizes the feature contributions to adjust for the greater number of ocular features. Overall, we see that the relative contributions of eye and vehicle data do not differ by interval duration. Features derived from ocular inputs are more influential in aggregate, but less influential on a per-feature basis.

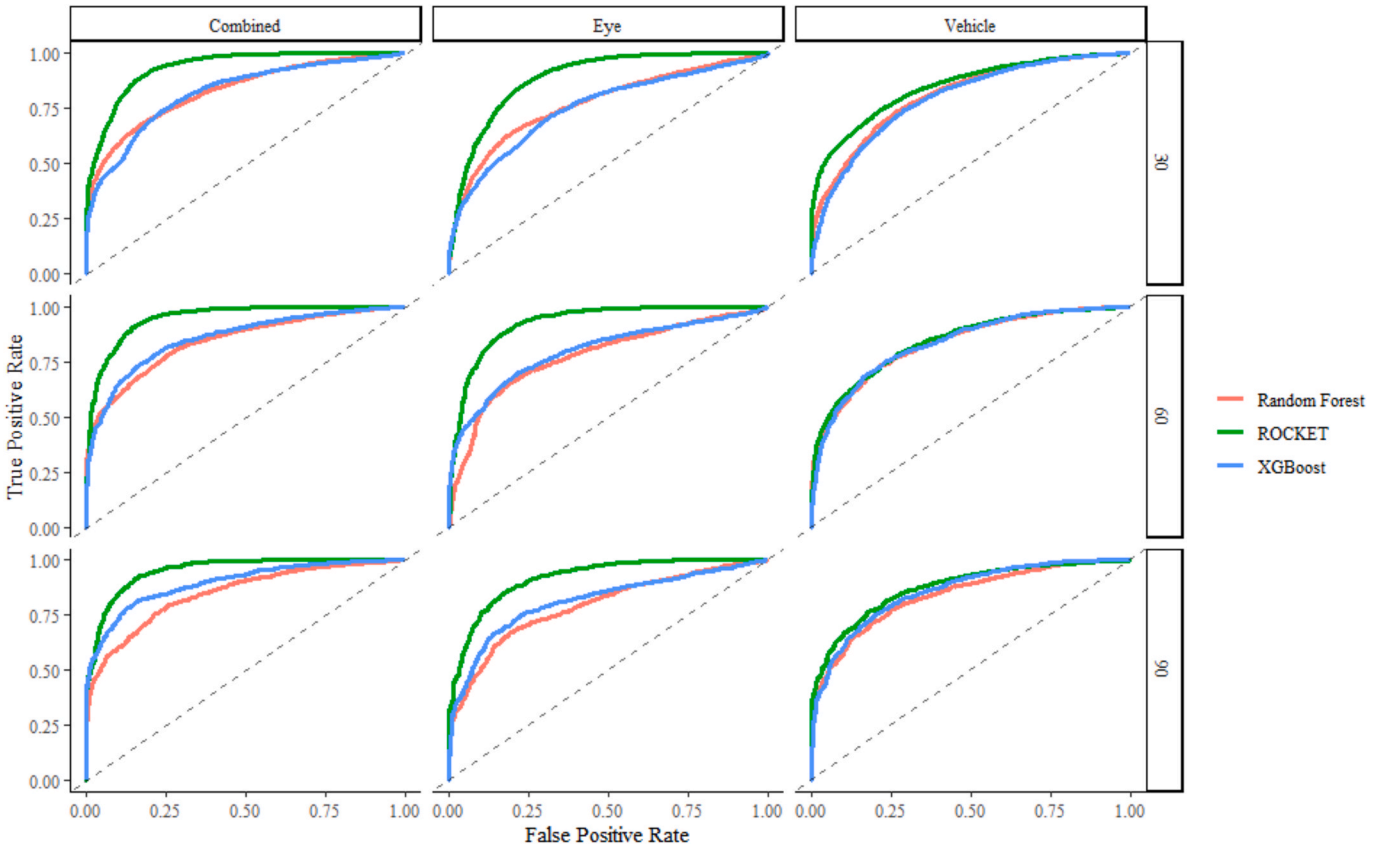


Fig. 2. ROC curves for select combinations of machine learning approach, data type, and interval duration (30, 60, and 90 s).

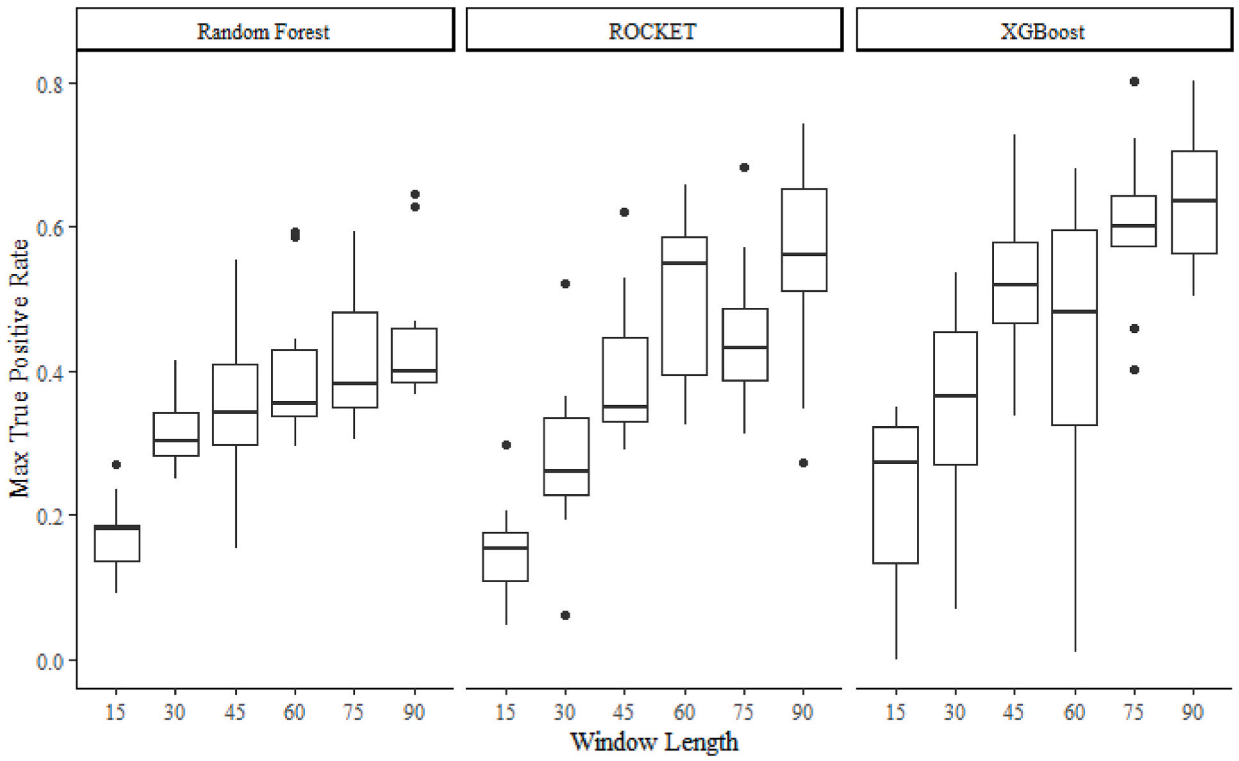


Fig. 3. True positive rate achieved at the prediction threshold producing exactly zero false positive classifications across 10 randomly assigned validation sets for each combination of modeling approach and interval duration using combined eye and vehicle data.

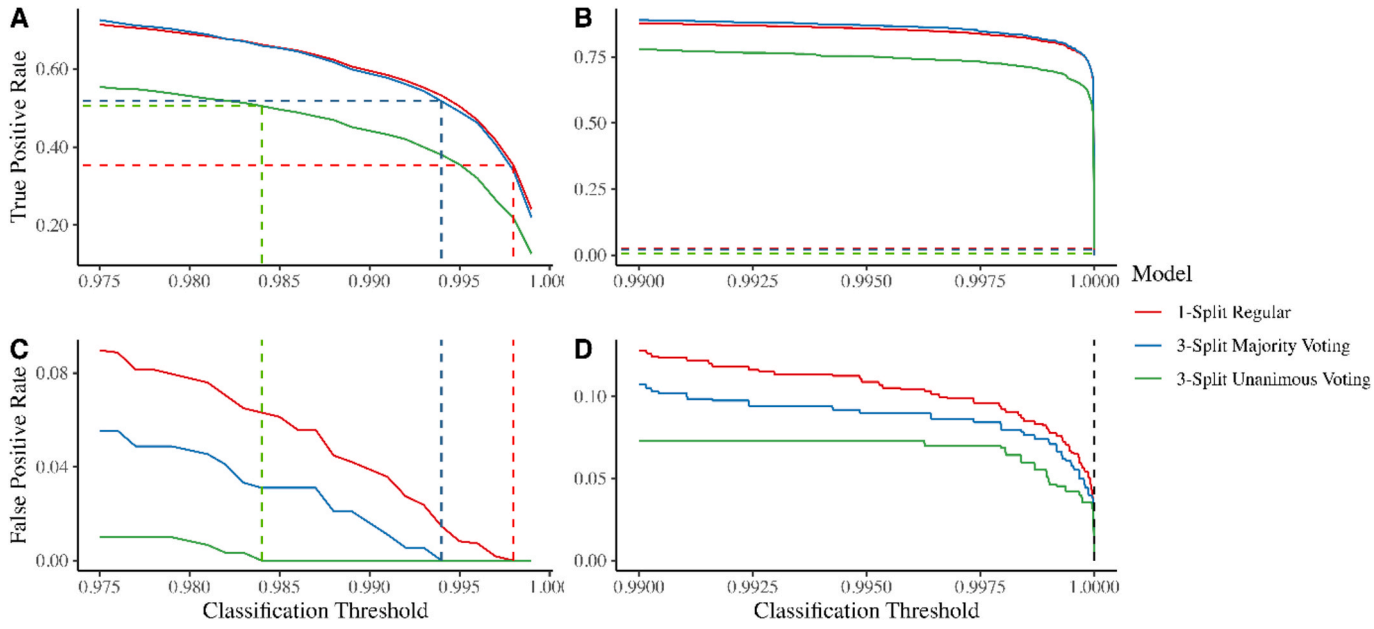


Fig. 4. TPR (top row) and FPR (bottom row) for XGBoost (left) and ROCKET (right) models trained on 90 s intervals under majority and unanimous voting schemes. Dashed lines indicate the classification threshold where exactly zero false positives are observed.

The raw and standardized contributions of feature subcategories (introduced in Table 1) are summarized in Table 4. The set of 21 features related to lateral vehicle control were most influential in aggregate, accounting for 25.3% of gains, and on a standardized basis (29.1%). Features capturing pupil and blink behavior were also very influential, accounting for 23.7% of gains in aggregate, and 23.9% when standardized.

Table 5 shows the top three most influential individual features for XGBoost models trained using each data type for 30, 60, and 90 s intervals. Consistent with the aggregate feature importance results, the most important individual features in models trained using both eye and vehicle data are related to pupil behavior and lateral control. Higher order moments, namely skewness and kurtosis, capture the most influential characteristics of the pupil size observed in an interval, while the standard deviation of the rate of steering angle change and lateral lane deviation tend to be the most influential vehicle-based characteristics.

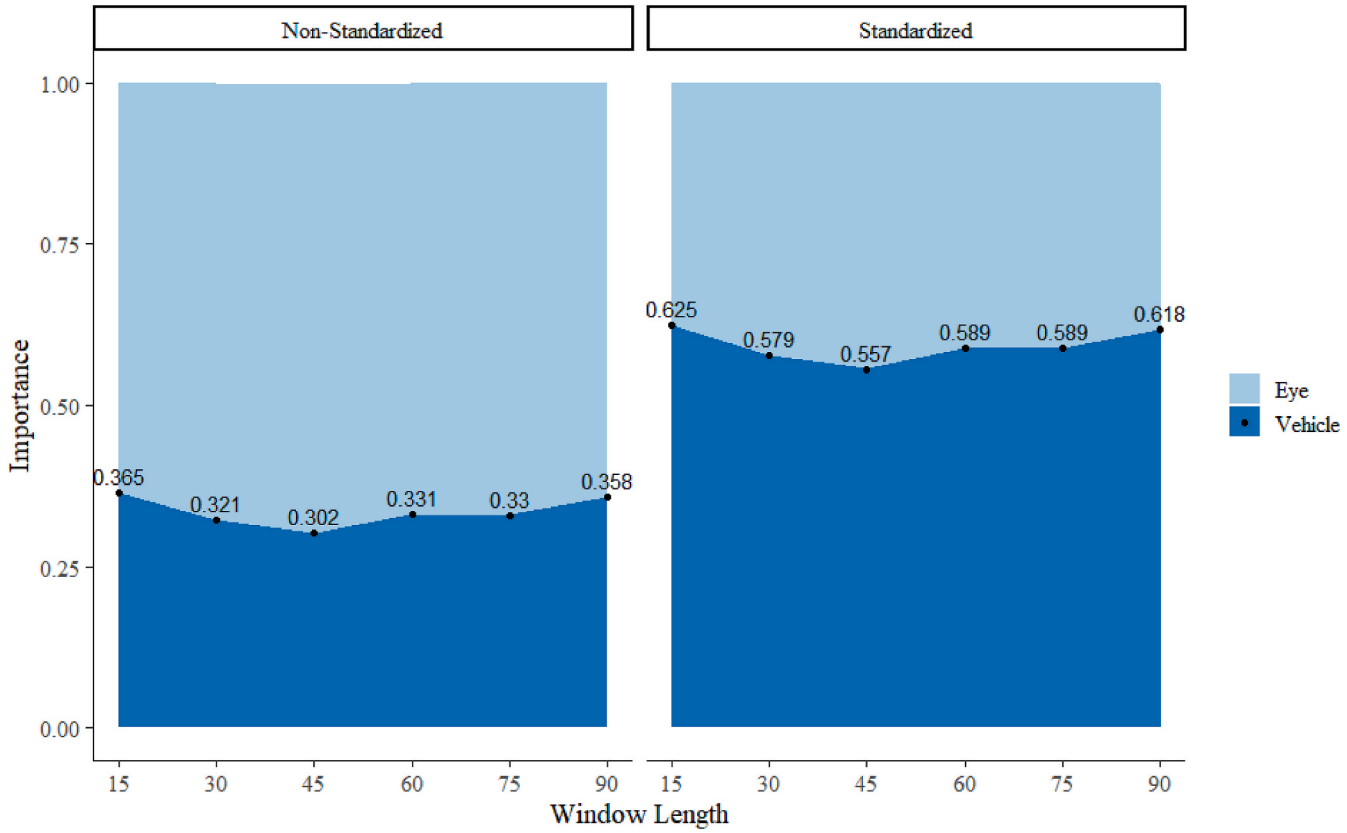


Fig. 5. Aggregate and standardized contributions of eye and vehicle related features using gain-based importance for XGBoost models.

Table 4
Aggregate and standardized contributions of feature subcategories using gain-based importance from XGBoost models trained on 90-second intervals of combined eye and vehicle data.

Feature Subcategory	Aggregate Contribution	Standardized Contribution	Number of Features
Lateral Vehicle Control	0.253	0.291	21
Pupil and Blink Behavior	0.237	0.239	24
Gaze Direction & Eye Movement	0.229	0.092	60
Head Position	0.128	0.147	21
Longitudinal Vehicle Control	0.100	0.127	19
Fixation Behavior	0.047	0.103	11

SHAP values for XGBoost and random forest models trained using 90-second intervals of combined vehicle and eye data are displayed in Fig. 6. The predictions of both models tend to be most influenced by similar features, with standard deviation of steering wheel angle rate being the most influential feature, followed by measures relating to pupil size, lane deviation, and blink duration. Notably, the 95th percentile of blink duration is the second most influential feature for XGBoost, while it was not one of the top three most influential features when using gain-based importance.

3.4. Robustness in new scenarios

Fig. 7 displays ROC curves of each modeling approach trained on 90-second intervals from the rural straight scenario and applied to 90-second intervals from the dark and gravel scenarios. The overall performance of models using eye-only and combined eye and vehicle data remains strong, with the top-performing models achieving AUC values exceeding 0.95. Vehicle-only results show some deterioration is performance, with the top models achieving AUC values of 0.87 and 0.74 for the gravel and dark scenarios, whereas the best vehicle-only model achieved a mean AUC of 0.88 across validation samples from the rural straight scenario.

4. Conclusions

As companies continue to develop and deploy camera-based driver monitoring systems it is important to evaluate the sensitivity and specificity of these technologies. Overall, we find that a combination of ocular and vehicle data measured over 90-second intervals provides sufficient information to identify alcohol presence with a high level of accuracy (AUC = 0.955). We also find that ocular and vehicle data collected by DMS provide complementary information in accurate identification of alcohol presence, with both data types providing distinct contributions and models trained using both types of data generally outperforming models trained using a single data type. However, we also found that when models trained on data from the rural straight scenario were applied to new driving environments there was essential no loss in performance observed when ocular features were used, while vehicle-only models did not carry over as well to these new environments. Altogether, this suggests that both inputs are valuable when seeking optimal predictions, but ocular features are more generalizable and robust to changes in the driving environment.

The optimal machine learning approach used to detect alcohol presence in drivers ultimately depends upon the underlying goals of the system. The convolutional-based method, ROCKET, yielded the best overall performance as measured by AUC in nearly all the scenarios we considered. This suggests the existence of predictive temporal patterns, particularly amongst ocular features, that we could not effectively capture using our handcrafted features. Future research should further

Table 5
Gain-based importance (mean and range) for the top three most influential features for XGBoost models trained using different types of input data for 30, 60, 90-second intervals.

Data Type	Interval (sec)	1st	2nd	3rd
Eye	30	skew_left_pupil_size 0.102 (0.050, 0.187)	kurt_left_pupil_size 0.069 (0.007, 0.156)	sd_left_pupil_size 0.032 (0.003, 0.159)
Eye	60	skew_left_pupil_size 0.075 (0.036, 0.120)	kurt_left_pupil_size 0.053 (0.003, 0.202)	kurt_right_pupil_size 0.029 (0.000, 0.161)
Eye	90	kurt_left_pupil_size 0.084 (0.001, 0.190)	skew_left_pupil_size 0.064 (0.030, 0.105)	sd_head_pos_y 0.045 (0.019, 0.101)
Vehicle	30	sd_steer_angle_rate 0.193 (0.113, 0.261)	middle_90_lane_dev 0.047 (0.028, 0.061)	q95_speed 0.045 (0.024, 0.073)
Vehicle	60	sd_steer_angle_rate 0.160 (0.100, 0.235)	sd_lane_dev 0.067 (0.022, 0.126)	sd_accel 0.050 (0.019, 0.102)
Vehicle	90	sd_steer_angle_rate 0.110 (0.053, 0.283)	sd_lane_dev 0.089 (0.028, 0.240)	sd_steer_angle 0.062 (0.008, 0.152)
Combined	30	skew_left_pupil_size 0.086 (0.032, 0.173)	sd_steer_angle_rate 0.051 (0.024, 0.077)	kurt_left_pupil_size 0.030 (0.002, 0.121)
Combined	60	skew_left_pupil_size 0.093 (0.027, 0.197)	sd_steer_angle_rate 0.042 (0.021, 0.063)	kurt_left_pupil_size 0.026 (0.004, 0.045)
Combined	90	skew_left_pupil_size 0.061 (0.018, 0.111)	sd_steer_angle_rate 0.049 (0.026, 0.095)	skew_right_pupil_size 0.029 (0.000, 0.173)

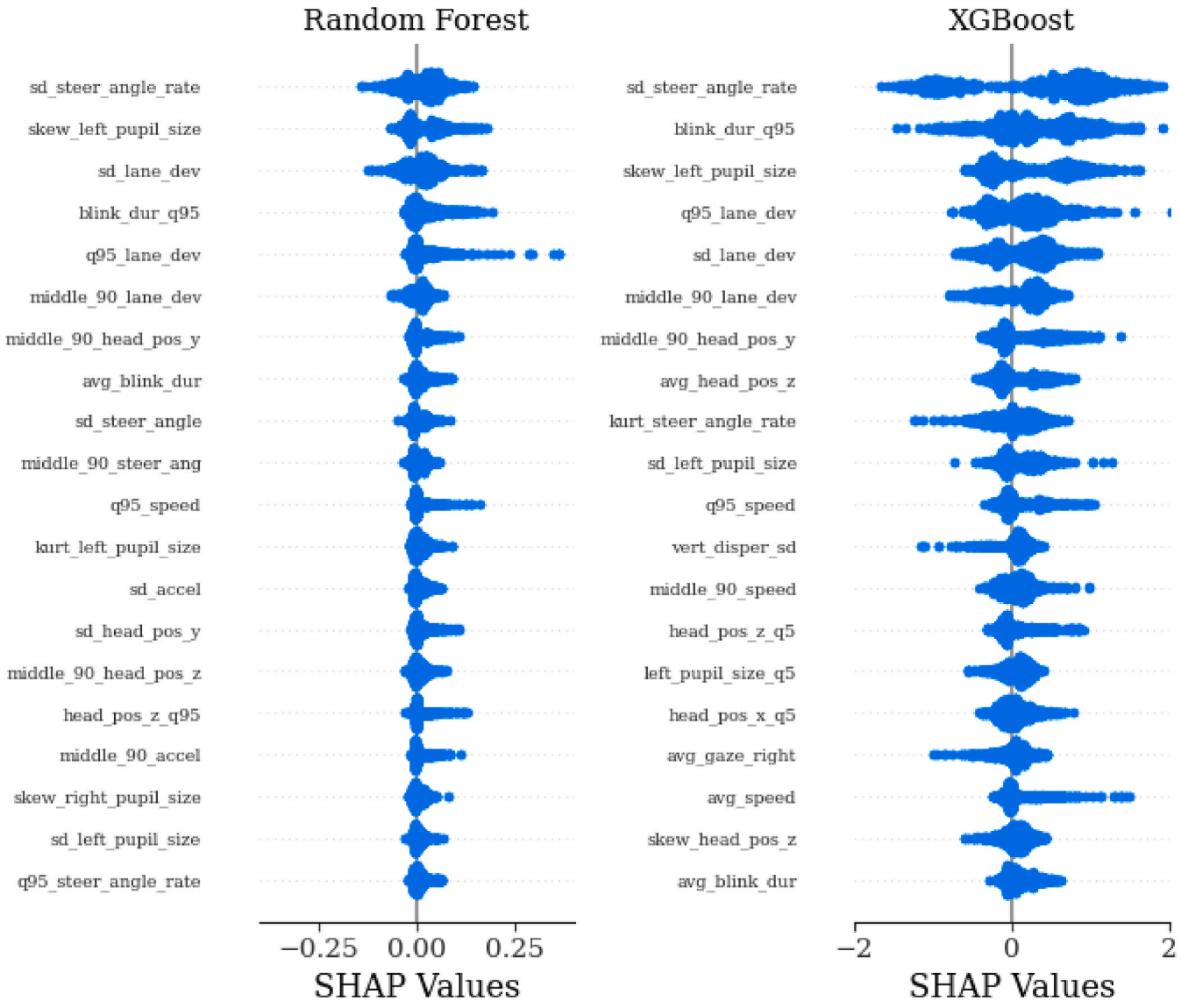


Fig. 6. SHAP values for XGBoost and random forest models trained on 90-second intervals of combined vehicle and eye data. Each point represents the impact of a feature on the prediction made for a single observation. Features are sorted by global importance, which is determined by the variance in SHAP values.

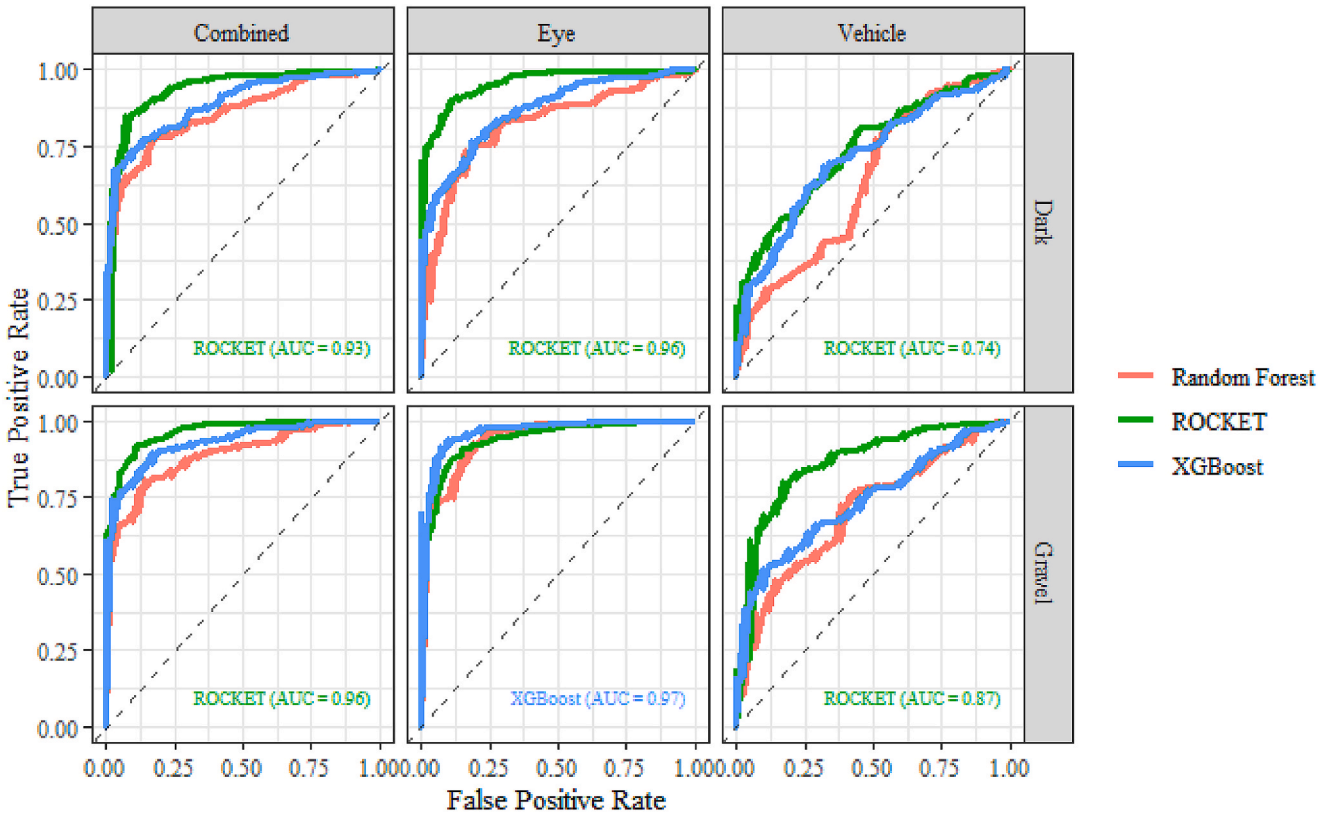


Fig. 7. ROC curves for each modeling approach trained on 90 s interval from the rural straight scenario and used to classify equal duration intervals from two new driving environments, dark and gravel. Each panel is annotated with the AUC of the best performing method for that scenario.

investigate the convolutional kernels used by ROCKET and the capacity for the patterns captured by these kernels to be encoded into a feature engineering approach. However, if a very low false positive rate is desired, as is the case in many DMS applications, XGBoost models trained using handcrafted features tend to perform as well or better than ROCKET models.

4.1. Limitations

One limitation of our analysis is the interplay between increasing levels of driver fatigue and decreasing BrAC in our training data. Some of our alcohol-positive samples were in dosing conditions with BrACs of 0.085 and 0.10 g/220 L with minimal fatigue, while others recorded later in the study were in dosing conditions with BrACs of 0.055 and 0.07 g/210 L with modest fatigue, and we did not have any alcohol-negative samples with heightened levels of fatigue or very low concentrations of alcohol. Consequently, our models may be picking up on signals of fatigue when identifying patterns associated with alcohol presence; however, in practice, scenarios involving lower BrAC and heightened levels of fatigue are well-aligned with alcohol consumption and driving patterns, reflecting the state under which an individual might drive home after a period of alcohol consumption.

The study sample is another limitation. Although efforts were made to recruit a diverse sample, the 36 healthy individuals who participated are not necessarily representative of the general population. Furthermore, many demographic subgroups were small, making it infeasible to compare model performance across these groups.

Other limitations include the use of a miniSim without advanced driver-assistance systems (ADAS), potentially leading to behavioral differences compared to real, on-road driving. While the miniSim has been validated for the detection of sensitive behavioral responses to impairment and distraction, and several studies have shown that ocular behaviors such as fixation duration and gaze position closely align in real road versus simulated environments, it is possible that subjects behave differently in a research environment where they are aware of the experimental stimuli and research objectives. Additionally, the simulated route exclusively used rural roadways, so it remains uncertain whether the eye and vehicle features identified and used in our models will be similarly predictive in urban and multi-lane highway settings.

4.2. Future research

Future research should consider a broader range of BrAC and fatigue combinations, particularly if a DMS aims to differentiate between alcohol-impaired and fatigued states. Additional research should also consider larger and more diverse samples to determine whether individual characteristics not represented in our data might influence DMS performance. For example, minor illnesses or neurological differences could potentially influence ocular metrics. Finally, future work should consider a breadth of driving environments, such as urban settings and multi-lane divided highways.

CRediT authorship contribution statement

Ryan Miller: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andy Chestovich:** Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Anshul Satav:** Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation. **Angela Eichelberger:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Jonny Kuo:** Writing – review & editing, Software, Resources. **Michael Lenne:** Writing – review & editing, Software, Resources. **Timothy Brown:** Writing – review & editing, Validation, Supervision, Resources, Project administration,

Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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